EXPERIMENT 40: TEST FOR ANTI-INFLAMMATORY AGENTS

40 STANDARD IN-COURSE MCQs

1. The major aim of Experiment 40 is to evaluate:

A. The local anesthetic potency of lidocaine B. The anti-inflammatory efficacy of pharmacological agents against acute paw edema in rats C. The antibiotic sensitivity of staphylococci D. The motor relaxant effect of benzodiazepines

Answer: B

2. What experimental animals are designated in the protocol on Page 183?

A. Adult guinea pigs B. Adult albino rats C. Albino mice D. New Zealand rabbits

Answer: B

3. What phlogistic (inflammation-inducing) agent is specified in the protocol?

A. Dilute hydrochloric acid B. Carrageenan (1% suspension) or fresh egg white C. Absolute ethanol D. Picrotoxin

Answer: B

4. What is carrageenan derived from chemically and biologically?

A. Snake venom toxin B. A natural sulfated polysaccharide extracted from red edible seaweeds (Chondrus crispus) C. Synthetic aromatic hydrocarbon D. Bacterial lipopolysaccharide endotoxin

Answer: B

5. Into which anatomical site of the rat is the phlogistic agent injected?

A. Tail vein B. Subplantar tissue (aponeurosis) of the right hind paw C. Anterior abdominal wall D. Dorsal neck scruff

Answer: B

6. What standard volume of phlogistic agent is injected subplantarly?

A. 0.01 mL B. 0.1 mL C. 1.0 mL D. 5.0 mL

Answer: B

7. What does the contralateral left hind paw receive as the internal vehicle control?

A. Equal volume of normal saline (0.1 mL) B. 1 mL of ethanol C. Nothing D. Carrageenan in double strength

Answer: A

8. What reference standard NSAID is specified on Page 183?

A. Morphine sulphate B. Acetylsalicylic acid (aspirin) or indomethacin C. Diazepam D. Phenobarbitone

Answer: B

9. What standard oral dose of acetylsalicylic acid (aspirin) is commonly used in this rat assay?

A. 1 mg/kg B. 100 mg/kg p.o. C. 1,000 mg/kg p.o. D. 0.1 mg/kg p.o.

Answer: B

10. When are the test agents and reference standards administered relative to carrageenan injection?

A. 1 hour prior (orally) or 30 minutes prior (i.p.) B. 24 hours after carrageenan C. In the same syringe mixed with carrageenan D. Only after the animal is sacrificed

Answer: A

11. What instrument is used to measure the thickness/diameter of the rat paw?

A. Sphygmomanometer B. Vernier caliper or micrometer gauge C. Glass measuring cylinder D. Analytical chemical balance

Answer: B

12. What advanced instrument measures paw edema by fluid (mercury or water) displacement?

A. Spirometer B. Plethysmometer C. Osmometer D. Spectrophotometer

Answer: B

13. Why are paw measurements taken in triplicate (3 repeated readings) for each animal?

A. Because the calipers stop working B. To minimize intra-observer measurement error and obtain a precise mathematical mean C. To stretch the rat's leg D. To allow carrageenan to absorb

Answer: B

14. At what initial time point is baseline paw diameter (D_0 or V_0) measured?

A. Immediately before carrageenan injection (0 hour) B. 24 hours after carrageenan C. At the end of the day only D. After the edema has completely resolved

Answer: A

15. At what post-injection time intervals is paw diameter subsequently monitored?

A. Once a day for 7 days B. Immediately after injection and at 30-minute intervals for up to 3 to 5 hours C. Every 10 seconds for 2 minutes D. Only at 24 hours

Answer: B

16. What mathematical formula is used to calculate the percentage inhibition of edema?

A. [(D_t - D_0)_control - (D_t - D_0)_treated] / (D_t - D_0)_control * 100 B. (D_t / D_0) * 100 C. [(D_t - D_0)_treated / (D_t - D_0)_control] * 100 D. (D_control + D_treated) / 100

Answer: A

17. If control rats have a mean edema increase of 2.0 mm and drug-treated rats have an increase of 0.5 mm, the % inhibition is:

A. 25% B. 50% C. 75% D. 90%

Answer: C

18. What graph is required to be plotted in Question 3 on Page 183?

A. Lineweaver-Burk plot B. Percentage inhibition against time (hours) C. Log dose versus mortality D. Body weight versus water intake

Answer: B

19. In Question 1 on Page 183, what are the five classical cardinal signs of acute inflammation?

A. Fever, chills, sweat, fatigue, nausea B. Rubor (redness), Calor (heat), Tumor (swelling), Dolor (pain), and Functio laesa (loss of function) C. Hypertension, tachycardia, tachypnea, polyuria, polydipsia D. Pallor, paresthesia, pulselessness, paralysis, poikilothermia

Answer: B

20. Which Roman encyclopedist first described the four classic signs (rubor, calor, tumor, dolor)?

A. Galen B. Aulus Cornelius Celsus C. Rudolf Virchow D. Hippocrates

Answer: B

21. Which 19th-century pathologist added the fifth sign, "functio laesa" (loss of function)?

A. Louis Pasteur B. Rudolf Virchow C. Robert Koch D. Paul Ehrlich

Answer: B

22. What vascular event is directly responsible for the "rubor" (redness) and "calor" (heat) of acute inflammation?

A. Local arteriolar vasodilation causing increased microvascular blood flow (active hyperemia) B. Massive vasoconstriction with capillary collapse C. Microvascular thrombosis and infarction D. Lymphatic obstruction

Answer: A

23. What vascular event is directly responsible for the "tumor" (swelling / edema)?

A. Decreased capillary hydrostatic pressure B. Increased microvascular permeability leading to plasma exudation into the interstitial tissue C. Influx of red blood cells into lymph nodes D. Proliferation of fibroblasts within 5 minutes

Answer: B

24. What chemical mediator sensitizes peripheral nociceptors (pain fibers) to produce "dolor" (pain)?

A. Prostaglandins (especially PGE_2 and PGI_2) and bradykinin B. Acetylcholine C. Insulin D. Thyroxine

Answer: A

25. The carrageenan-induced paw edema model exhibits what characteristic temporal pattern?

A. A monophasic instantaneous spike that resolves in 5 minutes B. A classic biphasic inflammatory response C. Zero inflammation for 24 hours followed by chronic fibrosis D. Continuous linear swelling lasting several weeks

Answer: B

26. The EARLY phase of carrageenan-induced edema (0 to 1.5 hours) is primarily mediated by:

A. Prostaglandins produced by COX-2 B. Release of histamine, 5-hydroxytryptamine (serotonin), and bradykinin from mast cells and platelets C. Infiltration of macrophages and lymphocytes D. Collagen synthesis

Answer: B

27. The LATE phase of carrageenan-induced edema (1.5 to 5 hours) is primarily mediated by:

A. Histamine only B. Induction of cyclooxygenase-2 (COX-2), synthesis of prostaglandins (PGE_2), and neutrophil infiltration C. Parasympathetic vagal discharge D. Release of stored iron from ferritin

Answer: B

28. NSAIDs such as aspirin and indomethacin primarily suppress which phase of carrageenan edema?

A. The late phase (1.5 to 5 hours) by blocking prostaglandin synthesis B. The early phase only (0 to 30 minutes) C. They produce zero effect on either phase D. They selectively stimulate the early phase

Answer: A

29. What is the fundamental molecular mechanism of action of nonsteroidal anti-inflammatory drugs (NSAIDs)?

A. Agonism of opioid mu receptors B. Inhibition of cyclooxygenase (COX-1 and COX-2) enzymes, preventing conversion of arachidonic acid to prostaglandins C. Blockade of histamine H_1 receptors D. Destruction of mast cell membranes

Answer: B

30. How does acetylsalicylic acid (aspirin) differ mechanistically from other NSAIDs (e.g., ibuprofen)?

A. Aspirin is a reversible competitive inhibitor B. Aspirin irreversibly acetylates a serine residue in the active catalytic site of COX-1 and COX-2 C. Aspirin directly degrades arachidonic acid D. Aspirin selectively activates lipoxygenase

Answer: B

31. Which cyclooxygenase isozyme is constitutively expressed and responsible for gastric mucosal protection and platelet thromboxane synthesis?

A. COX-1 B. COX-2 C. COX-3 D. Lipoxygenase

Answer: A

32. Which cyclooxygenase isozyme is predominantly inducible by inflammatory cytokines and mitogens at sites of tissue injury?

A. COX-1 B. COX-2 C. Cytochrome P450 D. Monoamine oxidase

Answer: B

33. How do glucocorticoids (e.g., dexamethasone, prednisolone) exert their anti-inflammatory actions?

A. Only by blocking histamine release B. Induce lipocortin-1 (annexin A1) to inhibit phospholipase A_2, and suppress transcription of COX-2, cytokines, and iNOS C. Directly stimulate the production of prostaglandins D. Inhibit renal excretion of uric acid

Answer: B

34. Why are glucocorticoids generally more broadly immunosuppressive and anti-inflammatory than NSAIDs?

A. They act on cell surface receptors only B. They inhibit upstream phospholipase A_2 (blocking both prostaglandins and leukotrienes) and repress multiple inflammatory genes C. They have no effect on arachidonic acid D. They destroy circulating neutrophils immediately

Answer: B

35. What is the primary cellular source of histamine in the early inflammatory response?

A. Red blood cells B. Tissue mast cells and circulating basophils C. Hepatocytes D. Skeletal muscle myocytes

Answer: B

36. Fresh egg white (albumin) induces paw edema primarily through:

A. Acute mast cell degranulation with anaphylactoid release of histamine and 5-HT B. Direct bacterial infection of the joint C. Mechanical compression of bone D. Destruction of the sciatic nerve

Answer: A

37. A potent antiserotonergic or antihistaminic drug administered prior to carrageenan would primarily suppress:

A. The late phase (after 3 hours) B. The early phase (0 to 1.5 hours) C. Both phases equally D. Neither phase

Answer: B

38. Which inflammatory peptide formed from plasma kininogens by kallikreins is a potent algogenic (pain-causing) and vasodilating mediator?

A. Angiotensin II B. Bradykinin C. Vasopressin D. Endothelin

Answer: B

39. In a preclinical research setting, what does a significant dose-dependent % inhibition of paw edema by an unknown plant extract prove?

A. The extract is an effective local anesthetic B. The extract possesses systemic acute anti-inflammatory activity C. The extract is highly toxic to rodents D. The extract destroys the central nervous system

Answer: B

40. The overall conclusion of Experiment 40 is that:

A. Paw edema is an uncontrolled process that cannot be inhibited B. The carrageenan-induced rat paw edema model is a validated, sensitive in vivo assay for evaluating acute anti-inflammatory drugs and differentiating early vs. late phase mediator mechanisms C. Inflammation in rats does not involve prostaglandins D. Only steroids can inhibit rodent paw edema

Answer: B